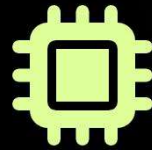


# Robo-soul H3.0

17-DOF Bipedal Humanoid Robot Technical Manual

A Comprehensive Guide to Mechanical Design, Actuation, and Kinematics

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# System Architecture

Hardware Foundations and Mechanical Configuration

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# Core Technical Specifications



## Physical Profile

Height: 390mm

Weight: 1.7kg (incl. battery)

Material: Aluminum Alloy (1mm brackets)



## Power System

Voltage: 7.4V Li-Po

Endurance: > 120 mins

Integrated low-voltage alarm protection



## Control Interfaces

PC Graphical Software

Android/iOS APP Control

Serial Communication (Secondary Dev)

# 17 Degrees of Freedom (DOF)

- ✓ **Head (1 DOF):** Rotation & Tilt
- ✓ **Shoulders (2 DOF):** Lateral movement
- ✓ **Arms (4 DOF):** Elbow & wrist flexion
- ✓ **Legs (8 DOF):** Hip, Knee, Ankle coordination
- ✓ **Feet (2 DOF):** Lateral stability and roll

*The high DOF count allows for complex maneuvers like backflips, lateral steps, and expressive dance moves.*

 Robo-soul H3.0 17DOF humanoid robot



# Digital Dynamics

High-Torque LDX-218 Servo Technology

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# Servo Torque Comparison (kg/cm)



LDX-218 digital servos provide the precision and torque necessary for lifting 1.7kg load while maintaining bipedal stability.

# LDX-218 Servo Features

 LDX-218 High Torque Servo Detail



## High Precision

Internal 32-bit MCU control with 0.3° resolution accuracy. Reduces jitter in complex poses.

## Mechanical Durability

Full metal gear train significantly extends service life and resists impact during falls.

# 32-Channel Controller

# 32

Simultaneous Servo PWM Channels

## STM32 Processing Core

- ✓ 16MB Action Storage (up to 10k groups)
  - ✓ Integrated MP3 module for audio sync
  - ✓ Plug-and-play wiring with anti-reverse
-



# Graphical Programming

## Visual Action Editing

The PC software allows users to edit "Action Groups" via a timeline interface. Each joint's angle can be adjusted precisely using sliders.

**Offline Execution:** Once groups are uploaded, the robot can execute sequences via controller buttons, APP commands, or hand-gesture sensors.

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# Connectivity & Control Methods

| Method        | Interface       | Primary Use Case                                 |
|---------------|-----------------|--------------------------------------------------|
| PC Software   | USB / Wireless  | Action sequence editing & initial calibration    |
| Mobile APP    | Bluetooth 4.0   | Remote joystick control & group triggers         |
| PS2 Wireless  | Controller Port | Real-time tactical control (Robotic Soccer)      |
| Secondary Dev | TTL Serial Port | Integration with Raspberry Pi / Arduino / Jetson |

# Motion Principles & Kinematics

Bipedal walking stability is achieved through the calculation of the Zero Moment Point (ZMP):

$$P_{zmp} = \frac{\sum m_i (\bar{z}_i + g) \bar{x}_i - \sum m_i \bar{x}_i \bar{z}_i}{\sum m_i (\bar{z}_i + g)}$$

- **Dynamic Balance:** Real-time center of gravity compensation.
  - **Inverse Kinematics:** Joint angle derivation based on foot-point coordinates.
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# Q&A

Thank you for your attention!

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For advanced support, visit our developer forum or contact technical support.

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# Image Sources



<https://images-cdn.ubuy.mv/635fc975cc51b91a4d76e4e3-assembled-aluminum-17dof-robo-soul-h3-0.jpg>

Source: [www.ubuy.mv](http://www.ubuy.mv)

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[http://ca.robotshop.com/cdn/shop/files/c66d4510\\_1\\_1200x1200.webp?v=1745576873](http://ca.robotshop.com/cdn/shop/files/c66d4510_1_1200x1200.webp?v=1745576873)

Source: [ca.robotshop.com](http://ca.robotshop.com)

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<https://www-robotics.jpl.nasa.gov/media/images/Interfaces5.original.png>

Source: [www-robotics.jpl.nasa.gov](http://www-robotics.jpl.nasa.gov)

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